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324812 (24)

BE (8th Semester)

Examination, April - May, 2012

Branch : Electrical, EEE

UTILIZATION OF ELECTRICAL ENERGY

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt any two questions from part (b), (c) and (d) of each question. Part (a) of each question is compulsory.

UNIT - I

Q. 1. (a) Name the corresponding quadrant operation, when an induction motor has been made to work as a brake. 2

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(b) Write advantages of electric drive. State its essential parts and explain it in brief. 7

(c) A motor drives two loads. One has rotational motion. It is coupled to the motor through a reduction gear with a = 0.1 and efficiency of 90%. The load has a moment of inertia of 10 kg-m^2 and torque of 10 N-m. Other load has translational motion and consists of 1000 kg weight to be lifted up at an uniform speed of 1.5 m/s. Coupling between this load and the motor has an efficiency of 85%. Motor has an inertia of 0.2 kg-m^2 and runs at a constant speed of 1420 r.p.m. Determine equivalent inertia referred to the motor shaft and power developed by the motor. 7 Q. 2.

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(d) What are the different components of load torque ? Describe each in brief. 7

UNIT - II

Q. 2. (a) What is maximum withstand temperature of class B insulation ? 2

(b) Explain phased locked loop (PLL) speed control scheme. 7

(c) The temperature rise of an electric motor is 40°C after an hour and 60°C after 2 hours. The motor current is 100 A. Determine approximately its final temperature rise when it works on load-cycle of 4 minutes working,

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8 minutes rest with a current of 125 A. Neglect the effects of iron losses. 7

(d) What are heating and cooling time constants ? Derive an expression for the temperature rise of an electric motor. 7

Q. 4

UNIT - III

Q. 3. (a) Give the name of different methods of starting of an induction motor. 2

(b) Describe static scherbius drive and explain it with a neat diagram. 7

(c) Explain current source inverter controlled three phase induction motor drive. 7

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(d) Explain with a suitable diagram the operation of a load commutated inverter fed synchronous-motor drive. 7

UNIT - IV

Q. 4. (a) Why d.c. series motor is used in traction system ? 2

(b) Explain the advantages of the semiconductor converter controlled drives. 7

(c) An electric train is to have acceleration and braking retardation of 0.8 km/h/s and 3.2 km/h/s respectively. If the ratio of maximum to

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average speed is 1.3 and time for stops 28 seconds, find schedule speed for a run of 1.5 km. Assume simplified trapzoidal speed-time curve. 7

(d) Describe the construction and operation of an electric arc furnace. 7

UNIT - V

Q. 5. (a) What is relation between Lux and Lumen per meter square ? 2

(b) What is street lighting ? Explain design considerations of street lighting. 7

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(c) What are the different energy conservation measures that are taken in electrical drives ? Explain in brief. 7

(d) A lamp having a uniform luminous intensity of 200 c.p. is suspended 7 meters above street level. What will be the illumination on the ground : 7

(i) Vertically beneath the lamp ?

(ii) 7 meters away from it ?

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